VANDANA DARAPANENI

Columbia, MO, USA • +1 (573) 219-0828 • vandana.d3883@gmail.com • LinkedIn

SUMMARY

Innovative Data Scientist with a master's in data science from the University of Missouri, specializing in data analysis, machine learning, business intelligence, and data engineering. Proficient in Python, R, SQL, and Java, with expertise in end-to-end model development, scalable data pipelines, and cloud-based deployment. Skilled in AWS, Docker, Airflow, and Git, optimizing system performance and automating workflows. Adept at transforming complex data into actionable insights while ensuring security, scalability, and efficiency. A quick learner with a growth mindset, I thrive in dynamic environments and adapt seamlessly to new technologies and challenges. I am a strategic problem-solver, passionate about delivering impactful AI solutions and collaborating with stakeholders to drive data-driven decision-making and business growth.

EDUCATION

University of Missouri**,** Columbia, MO, USA — Aug 2022 – May 2024

Master of Computer Science, GPA: 3.9/4.0

- ***Relevant Coursework:*** *Introduction to Data Science, Big Data Visualization, Data Mining and information retrieval, Data Analytics from Applied Machine Learning, Parallel Computing, Big Data Security, Natural Language Processing, Statistics & Mathematics, Database and Analytics.*

Siddartha Institute of Science & Technology Puttur**,** AP, INDIA — Aug 2017 – Jun 2021

Bachelor of Technology in Computer Science, GPA: 3.6/4.0

- ***Relevant Coursework:*** *Data Structures and Algorithms, Computer Organization, Data Analytics, Machine Learning, Pattern Recognition, Data Base Management System, Neural Networks, Map Reduce, Computing Systems for Data Processing, software engineering.*

SKILLS & OTHER

Programming Languages: Python, R, Java, C, C++.
Data Wrangling: Pandas, NumPy, Dask, Pelt, dplyr, tidyverse.
Data Visualization: Matplotlib, Seaborn, Plotly, Sweetviz, ggplot.
Databases: MySQL, Microsoft SQL, Postgres SQL, NoSQL, RDBMS.
Statistical Analysis: SciPy, Statmodels, SAS, Parametric Methods, Non-Parametric Methods, Distribution-free test, Post-Hoc Analysis, Correlation Tests.
Machine Learning: TensorFlow, PyTorch, Scikit-Learn, Keras, pickle file, supervised learning, unsupervised learning, reinforcement learning, ensemble learning, deep learning, classification algorithms, regression algorithms.
Data Engineering: dbt, Airflow, Kafka, Snowflake, Fivetran, Hadoop, Databricks.
Remote Sensing GIS: ArcGIS, QGIS, Planet, Sentinel Hub, Landsat, Geojson.io, GEE
Natural Language Processing: NLTK, Spacy, Transformers (Hugging Face), Generative AI, large language models.
Tools & Platforms: AWS (SageMaker, Lambda, S3, EC2, EKS, API Gateway), GCP, Azure, Git version control, Docker, Kubernetes, Bash, Power BI, Tableau, MS Excel, Micrsoft Office 365, JIRA, Agile, jupyter notebook, pycharm, VSCode, R studio.
Areas of Expertise: Data Science Life Cycle, Statisctial Analytics, Descriptive Analysis, Exploratory Data Analysis, Machine Learning models development and deployment, Data Structure and Algorithms, Data Visualization, BI Tools, ETL pipelines, containerization, orchestration and geospatial analysis.

PROFESSIONAL EXPERIENCE

University of Missouri (Agriculture Engineering), Columbia, MO, USA — Sep 2023 – Present

Junior Data Scientist, Full Stack

- Collected high-resolution spectral imagery and weather data from Planet Lab and local stations, using Python API and Selenium for data acquisition.
- Managed and processed data with MSSQL, Excel, and Python to estimate pasture biomass by integrating remote sensing, ground truth, and weather data.
- Integrated ground truth data with vegetation indices and performed data wrangling using Apache Airflow, Pandas, and NumPy for model readiness.
- Designed interactive Power BI and Excel dashboards, improving data accessibility by 25%.
- Trained machine learning models (Random Forest, Ridge Regression) on 6 years of data, achieving 10% accuracy improvement and optimizing model precision.
- Conducted statistical analysis using SAS and Python, revealing significant treatment effects ($p = 0.019$) and cross-validated models on pilot farms, maintaining 83% accuracy.
- Deployed models to production using Git, Docker, React, and Fast API, implemented ETL pipelines with Airflow and Snowflake, and developed a neural network to predict pasture biomass based on GPS tracking and satellite imagery.

Stock Group Analytics, Great Falls, VA, USA — Nov 2024 – Present

Data Analyst

- Assisted with the project, phishing attacks received from various sources, ensuring accuracy and completeness for effective analysis by leveraging data preprocessing python libraries
- Applied statistical methods using python stats libraries and tools to discover hidden patterns of datasets, uncover trends, and deliver actionable insights.
- Developed detailed reports and visualizations using power BI and Python data visualization libraries to effectively communicate findings to stakeholders.
- Supported ongoing data analytics projects, contributing to the success of key company initiatives.

University of Missouri (Animal Science), Columbia, MO, USA — Jun 2023 – Aug 2023

Data Analyst

- Maintained 5 years of large-scale soybean samples and quality datasets using Microsoft Excel and snowflake improve the accessibility of data by 70%.
- Conducted in-depth exploratory data analysis (EDA) using DAX in Power BI, revealing a statistically significant correlation between soybean protein levels and improved growth rates in poultry, providing actionable insights for feed optimization.
- Created engaging visualizations with Python and Tableau, turning data into intuitive notebooks that made it easy for stakeholders to grasp important trends and make confident, informed decisions increase communication efficiency by 65%.

University of Missouri (Psychology Department), Columbia, MO, USA — Oct 2022 – June 2023

Graduate Research Assistant

- Contributed to advancing research in cognitive psychology by developing a specialized memorization task utilizing random consonant generation and machine learning models.
- Created Power BI reports to analyze working memory dynamics in adolescents, enhancing data interpretation for researchers.
- Optimized existing dashboards for improved performance and usability, reducing load times by 40%

Star Health Insurance, Hyderabad, India — July 2021 – June 2022

Data Analyst

- Extracted the insurance data along with patient records and policy claims using MySQL from the database to perform the data cleaning using Pandas and NumPy for conducting further analysis.

- Worked in Automating the data extraction and reporting tasks using python to reduce the manual reporting time by 20%.
- Collaborated with cross functional teams to provide data driven insights that explain policy pricing, risk management and patient care strategies.
- Performed Statistical Analysis on insurance claims to identify cost driers, risk profiles and fraud claims, resulting in a 15% reduction in claims errors.
- Supported senior data scientists for building predictive models to forecast healthcare utilization and claims frequency using python and SQL.

Kea Info Tech, Andhra Pradesh, India — Sep 2019 – Dec 2019

Software Intern

- Worked in a geo spatial analysis project using Python and ML algorithms to process remote sensing data collected from satellite images in Google Earth Engine, classifying areas into Forests, barren lands, and agricultural zones, enabling real-time analysis of decade-long satellite data.

PROJECTS

Predicting The Risk of Coronary Heart Disease In USA

- Utilized full stack Data science skills from extraction of data to build predictive model, achieving 78% accuracy. Designed an infographic to communicate key findings and support data-driven health interventions.

Analyzing Economic Sentiment from Financial News for Market Prediction

- Applied tokenization and sentiment analysis to assess the correlation between financial news and stock trends, analyzed tweets and financial reports for sentiment insights, built predictive models (ARIMA, GARCH) achieving an R^2 score of 0.73.

Predicting Treatment Delay Period for Metastatic Breast Cancer

- Cleaned and preprocessed data using Python (Pandas, NumPy), performed exploratory analysis with Matplotlib and Seaborn, optimized the ETL pipeline using Airflow, and built predictive models (Random Forest, LightGBM, CatBoost), achieving 84% accuracy.

Predicting Flight Delays using Machine Leaning

- Led the project utilizing supervised machine learning (Decision Trees, Gradient Boosting) with an 89% accuracy rate. Integrated the predictive model into a React and Angular front-end, allowing users to access real-time delay predictions.

E-Commerce Data Analysis

- Analyzed data to identify a high return rate on an e-commerce platform for a clothing store in New York, providing a conclusion to invest more on application than website.

Global Airline Accessibility

- Visualized the analysis of global airline accessibility in the year 2012 utilizing the data visualization python libraries.

Sentiment Analysis using GCP and Kubernetes

- Gained Experience in utilizing the Google Cloud Platform (GCP) resources to conduct web scraping on the Reddit website, followed by sentiment analysis on the obtained data. The findings revealed a higher prevalence of comments falling within the neutral and negative categories as opposed to the positive category.

VOLUNTEER WORK

Geospatial analysis for finding deforestation in Missouri

- Utilized Google Earth Engine (GEE) and Python to analyze and process land cover changes over the past 15 to 20 years, specifically identifying forested areas that have transitioned into barren land due to deforestation, with a focus on land shape transformation.

Dashboard Creation for nutrition data

- Worked with Power BI and Tableau to create an interactive dashboard explaining the types of nutrition values of grains and millets along with the benefits of consumptions of healthy grains.

Image segmentation analysis on corn field

- Utilized computer vision YOLO concept for detecting the corn fields from the UV radar images collected in fields and performed clustering-based image segmentation.

Machine Learning for transcriptomics

- Worked on plant transcriptomics data to apply supervised machine learning algorithms to identify the species and region of the plant.

Bloomberg Data Scrapping

- Utilized the Bloomberg station available in the university library to collect the carry data for equity on a monthly basis which has been further used to perform time series analysis on the sales of stocks market.

CERTIFICATIONS

- **Certified SQL Intermediate Level from Q-Spiders.**
- **Data Science with Python from Perfect eLearning.**
- **Machine Learning with Python from Perfect eLearning.**
- **Internet Of Things from APSSDC.**